

# Employee Wellness Recommendation System

*Using Content-Based and Cluster-Based Hybrid Recommenders, with Offline Ranking Evaluation and Cold-Start Mitigation*

## METHODS & TOPICS COVERED

- 01

**Content-Based Filtering**

Cosine k-NN similarity
- 03

**Ranking Evaluation**

Precision@K · MAP@K · NDCG@K

- 02

**Cluster-Based Hybrid**

KMeans + collaborative CF
- 04

**Cold-Start & Scalability**

Two-tier mitigation strategy

## PREPARED BY

Taimoor Ahmad  
Mohd Daneel Khan  
Usama Murtaza Awan

20054191  
20084451  
20060983

## MODULE LECTURER

Nitya Govindaraju

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## **PART 1: INTRODUCTION**

### **1.1 Business Objectives of the Project**

- Enable HR teams to monitor employee wellness trends.
- Improve workforce productivity by identifying factors affecting employee well-being.
- Reduce employee burnout through early identification of risky employees.
- Support evidence-based decision-making by providing managers with personalised insights into employee wellness and engagement.

### **1.2 Introduction of the Project**

Employee well-being has become a central concern for modern organisations because poor mental health, insufficient sleep, excessive workloads, and burnout lead to reduced productivity. Human Resources teams need data-driven tools that can identify vulnerable employees. Such tools can suggest targeted interventions rather than one-size-fits-all wellness programs. The development and testing of a system that recommends employee wellness activities. The system fetches the most similar employees with respect to well-being profile (sleep, workload, exercise, mental well-being, job satisfaction, burnout) of the provided query employee, surfaces the personalized wellness interventions in conjunction with a clear rule-based layer.

Two different recommendation paradigms are presented and evaluated in accordance with a common evaluation protocol:

- Content Based Filtering (CBF): cosine-similarity k-Nearest-Neighbours model over an engineered wellness feature space with a search of the entire training population per query.
- Cluster-Based Hybrid Recommender: unsupervised KMeans clustering in the PCA-reduced feature space and then CF in the cluster of the user who submitted a query.

The project also introduces a full offline assessment via Precision@K, MAP@K and NDCG@K, while using a scalability as a structural side-benefit of the cluster-first design, a cold-start mitigation strategy is implemented. A Generative AI assessment is implemented, where a Variational Autoencoder (VAE) is implemented in PyTorch and generates synthetic employee profile inputs that are directly fed into the cold-start pathway.

### 1.3. Dataset Description

It is built on the Remote Work Productivity dataset consisting of 50,000 employees from 15 attributes. One employee record captures information about the employee's demographics, working arrangement, and wellness indicators as shown in Table 1.

*Table 1: Description of Dataset*

| Attribute                   | Type        | Description                                                                         |
|-----------------------------|-------------|-------------------------------------------------------------------------------------|
| employee_id                 | Identifier  | Unique employee identifier (EMP000001 to EMP050000)                                 |
| age, country, industry      | Demographic | Age 21-59; 8 countries; 5 industries (Healthcare, Tech, Education, Finance, Retail) |
| work_mode                   | Categorical | On-site, Remote, or Hybrid (approximately balanced, ~16.6k each)                    |
| year                        | Numeric     | Observation year, 2018-2025                                                         |
| weekly_hours_worked         | Numeric     | 30-60 hours per week                                                                |
| tasks_completed             | Numeric     | Task throughput proxy                                                               |
| productivity_score          | Numeric     | 0-100 productivity outcome (used only as ground truth)                              |
| sleep_hours_avg             | Numeric     | Average nightly sleep, 4-9 hours                                                    |
| exercise_frequency_per_week | Numeric     | 0-6 sessions per week                                                               |
| mental_health_score         | Numeric     | 0-100 mental health indicator                                                       |
| burnout_level               | Categorical | Low / Medium / High                                                                 |
| job_satisfaction            | Numeric     | Satisfaction score                                                                  |
| salary_usd                  | Numeric     | Annual salary in USD                                                                |

This dataset is appropriate for a recommender-systems formulation for two reasons. First, the wellness attributes are a natural item/user content area where there is a meaningful similarity between employees. Second, `productivity_score` is an external outcome variable that could serve as the criterion for which to determine the relevance of the recommendations without contaminating the similarity features, which allows for non-circular offline evaluation.

The distributions of the key numeric wellness features after cleaning are shown in Figure 1. The variables of productivity, mental health and job satisfaction are left-skewed with higher values (generally healthy workforce), and the variables of sleep, weekly hours and tasks completed are near symmetrical.



Figure 1: Distributions of the six numeric wellness features used by the recommenders.

The categorical features are well balanced, which is good for clustering and encoding. The split of On-site, Remote and Hybrid work mode is almost even, while industries are very even, and Low burnout level is more prevalent than High and Medium as shown in the Figure 2.

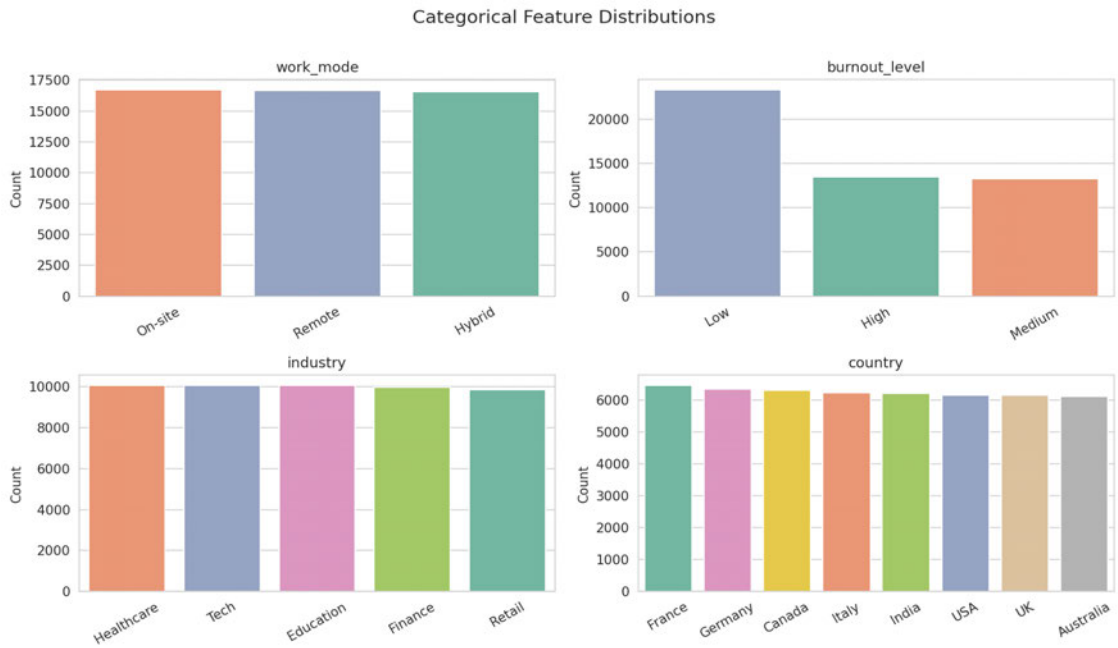


Figure 2: Displays for the categorical features: work mode, burnout level, industry, country.

## **PART 2: SYSTEM DESIGN AND CORE IMPLEMENTATION**

### **2.1 Data Pre-Processing and Feature Engineering**

#### **2.1.1 Data Cleaning**

The data set was checked for missing values and potential duplications, with no such gaps or repetitions found for all 50,000 records - a median-imputation fallback is however implemented for robustness on future data. Outliers were detected using IQR based outlier audit ( $1.5 \times \text{IQR}$  fences) and the number of outliers found in each column is indicated below: tasks\_completed (3), productivity\_score (728), mental\_health\_score (964), job\_satisfaction (1,197).

Outlier treatment uses a copy of the dataframe, so that no data is dropped from the sample; the original sample is recommended. Treatment strategy was automatically selected per column based on skewness; if  $|\text{skewness}| > 1$  then the column was treated by capping at the fences (as shown in the table, this was done for the column productivity\_score and mental\_health\_score); if approximately symmetric, then Winsorising was performed at the same fences (as shown in the table this was done for the column tasks\_completed and job\_satisfaction). Since the extreme tails of job\_satisfaction contain little signal, it was fine to have it pass a second time with a tighter 5th-95th percentile to preserve its semantics; likewise, score columns with value domains that were not integers were rounded after being capped. All ten numeric columns were cleaned successfully with a recheck.

#### **2.1.2 Exploratory Data Analysis and Derived Recommendation Rules**

To determine those features that have the most influence on productivity, univariate, bivariate and multivariate analyses were conducted to identify these features in the recommender feature space. The correlation heatmap (Figure 3) shows the strongest positive correlations with productivity\_score are mental\_health\_score (+0.72), sleep\_hours\_avg (+0.58), tasks\_completed (+0.55), and job\_satisfaction (+0.51), with weekly\_hours\_worked showing a strong negative relationship (-0.56). There is virtually no linear correlation between productivity and demographic and pay variables including age, salary, year, and frequency of exercise.

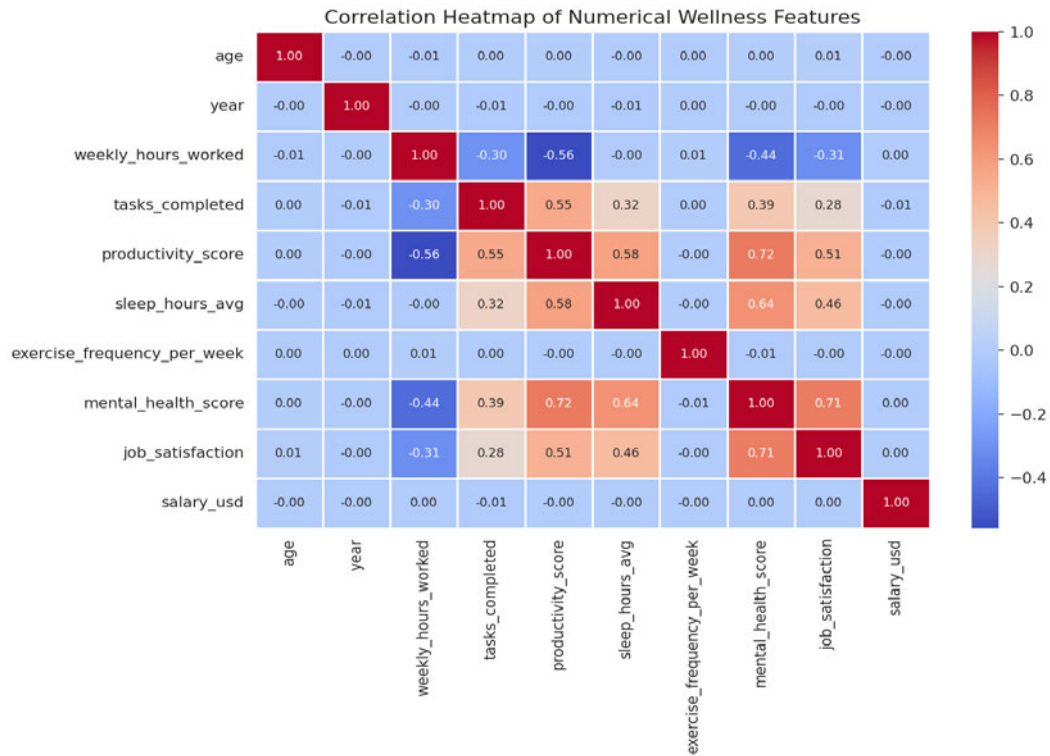


Figure 3: Correlation Heatmap of all numeric features.

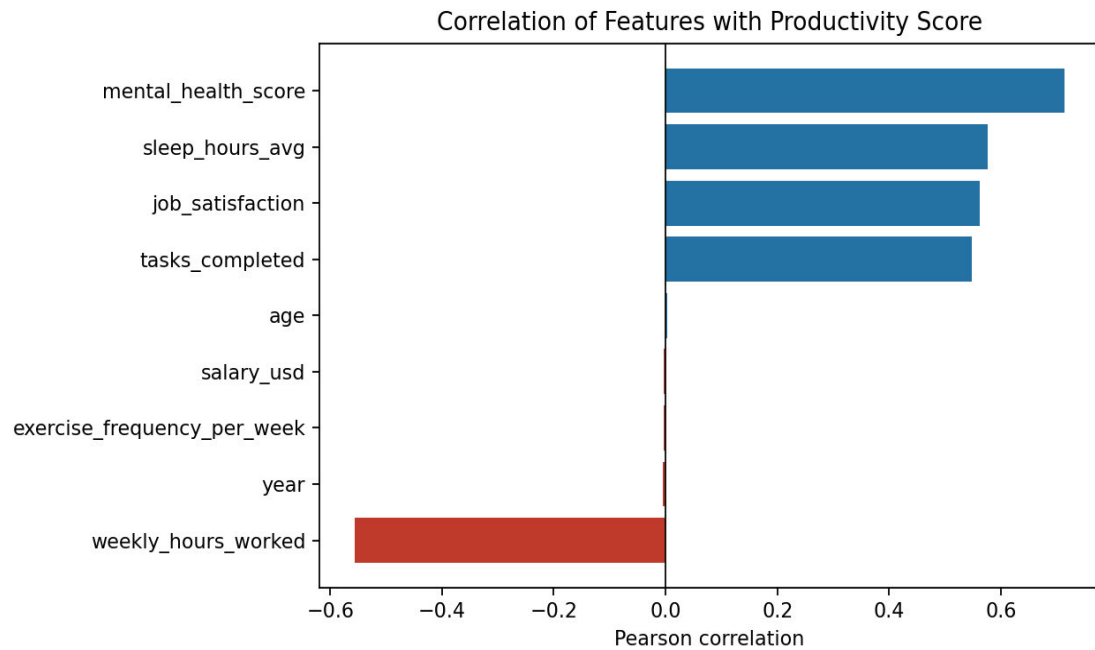


Figure 4: Ranked correlation of each numeric feature with productivity\_score.

These trends were also confirmed in the bivariate analyses. The good sleepers have higher and stable productivity, whereas low sleepers have a much larger range as shown in Figure 5.

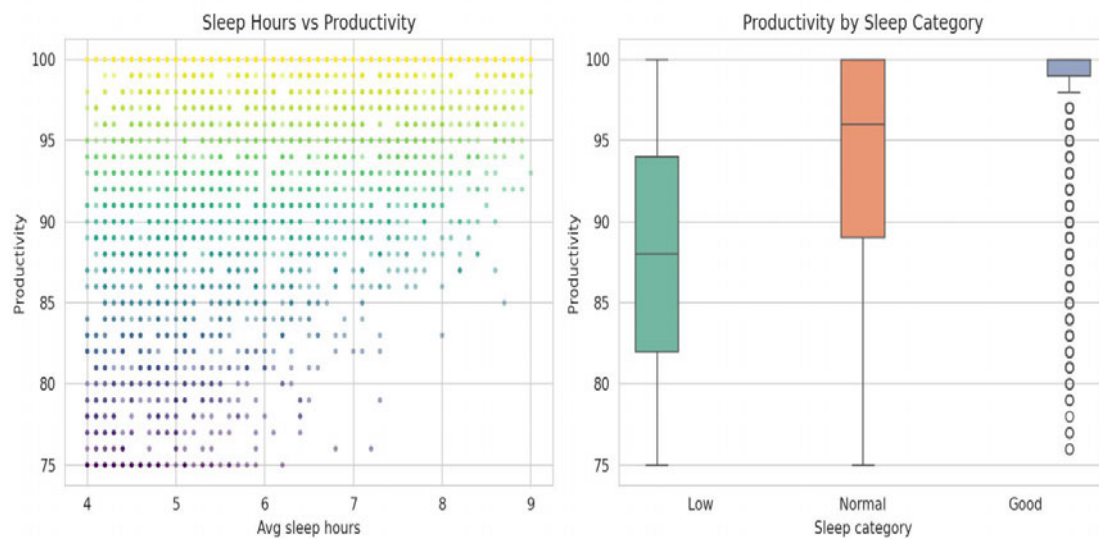


Figure 5: Sleep hours versus productivity (scatter) and productivity by sleep category (boxplot).

Likewise, low exercisers have lower mental health scores and high-burnout employees have the longest average of working weeks (Figure 6); which leads to two of the rule based recommendations.

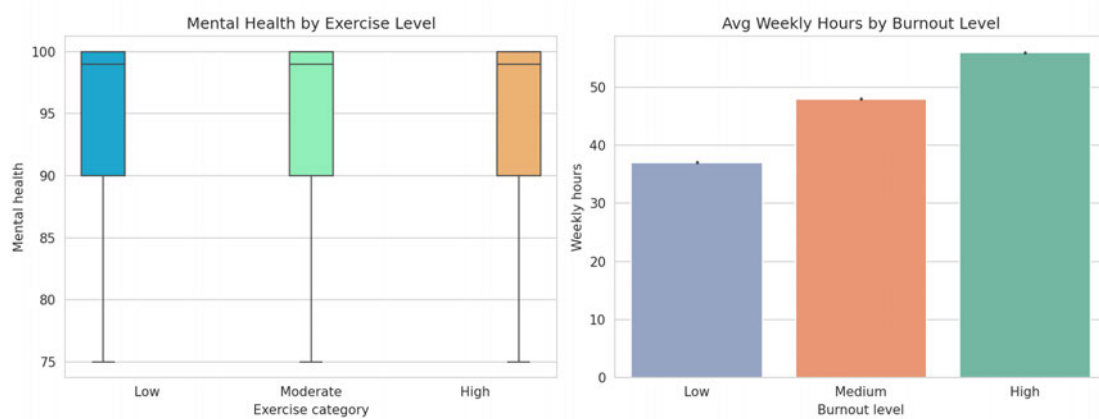


Figure 6: Mental health by exercise level, and average weekly hours by burnout level.

Higher mental health and higher job satisfaction are each informative of higher productivity and are correlated with one another in Figure 7, supporting the argument that mental health is a well-normative indicator of wellness that tracks job satisfaction.

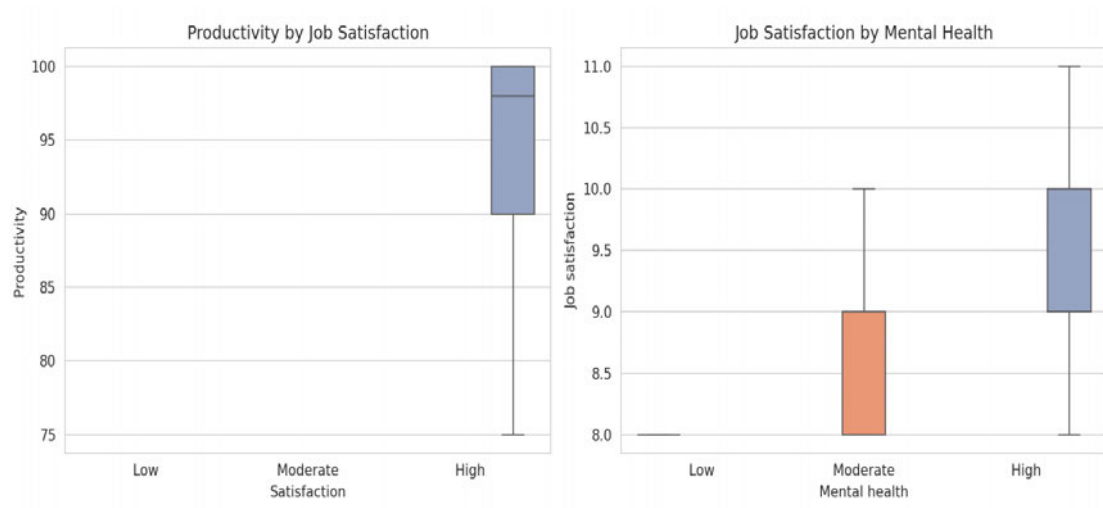


Figure 7: Productivity by job satisfaction, and job satisfaction by mental health level.

When broken down by multivariate country-wise and age-wise, the differences are negligible (high proportion of people in burnout ranging from 26% to 28% across the eight countries and mean productivity from 94.36 to 94.52), demonstrating that behavioural wellness variables, rather than demographics, are the most significant factors in Figure 8. This made it valid not to include country, age and salary as recommender features.

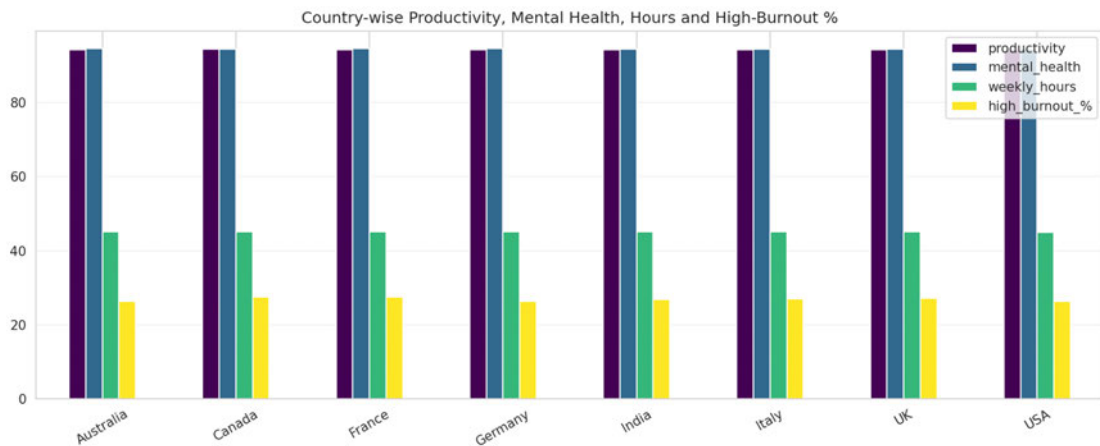


Figure 8: Country-wise productivity, mental health, weekly hours, and high-burnout percentage.

The EDA also resulted in four transparent IF-THEN rules (e.g. IF sleep is low AND productivity is below the peer median THEN recommend improving sleep routine) that were rolled into a rule-based baseline recommender. This baseline is not a ranking system, it is not rated using ranking metrics, it is only a sanity check to interpret and it could be an explainable advice layer in an HR dashboard with the ML recommendations.

## 2.2 One Canonical Feature Pipeline

Both recommenders have exactly the same feature representation, which is created using a scikit-learn ColumnTransformer; thus, the evaluation results are comparable and the fitted pipeline could be reused when evaluating a so far unseen recommender with new employees at inference time. All 50,000 employees have all six numeric features (weekly\_hours\_worked, tasks\_completed, sleep\_hours\_avg, exercise\_frequency\_per\_week, mental\_health\_score, job\_satisfaction) MinMax-scaled and also all two categorical features (burnout\_level, work\_mode) are one-hot encoded with the first level dropped resulting in a 10-dimensional feature matrix.

## 2.3 Recommender 1: Content-Based Filtering (CBF)

The content-based recommender is trained on the training pool only, and uses a cosine-similarity NearestNeighbors model (brute-force search). The profile of the query employee is projected onto the common feature space (vector projection), and the  $k$  most similar employees among the training-pool are returned together with the similarity score (1 minus cosine distance) of query employee and the returned employee (1 minus cosine distance). Cosine similarity was selected instead of Euclidean distance, since after MinMax scaling, it is more suitable to highlight the shape of a wellness profile (the proportion between sleep, workload and satisfaction) than the absolute values, which would represent the appropriate notion of similar wellness situation. The motivation for the first paradigm is that the problem has a lot of item content (all the employees are fully described with wellness attributes) but no classical user-item interaction matrix, so that a pure rating-based collaborative filtering wouldn't work. CBF does not need any history of interaction, is very forgiving to new employees, and recommendations are directly explainable in terms of matching features.

## 2.4 Recommender 2: KMeans Cluster-Based Hybrid

The second, structurally different recommender is based on both unsupervised and neighbourhood-based collaborative filtering. First, PCA is applied only to the training pool, and the number of components to retain is based on the 90% variance rule, not on a fixed number; this decreased the number of features from 10 to 6. Second, KMeans clustering is applied in the PCA space where  $K$  is chosen by a systematic search indicating  $K$  varies from 2 to 19 and four criteria are used to assess the results, namely inertia (elbow method), silhouette score, Davies-Bouldin index, and Calinski-Harabasz index.

Table 2: Choosing  $K$  (elbow + Silhouette / Davies-Bouldin / Calinski-Harabasz)

| <b>K</b> | <b>Inertia</b>  | <b>Silhouette Score</b> | <b>Davies–Bouldin Index (DBI)</b> | <b>Calinski–Harabasz Index (CHI)</b> |
|----------|-----------------|-------------------------|-----------------------------------|--------------------------------------|
| 2        | 35,531.8        | 0.3030                  | 1.4610                            | 17,678.45                            |
| 3        | 29,589.2        | 0.2987                  | 1.4533                            | 14,630.63                            |
| 4        | 24,423.8        | 0.3170                  | 1.3351                            | 14,635.86                            |
| 5        | 20,481.1        | 0.3474                  | 1.2470                            | 15,014.51                            |
| 6        | 17,381.0        | 0.3830                  | 1.1557                            | 15,580.32                            |
| 7        | 15,386.9        | 0.3906                  | 1.0655                            | 15,529.76                            |
| 8        | 13,618.9        | 0.4161                  | 1.0491                            | 15,780.46                            |
| <b>9</b> | <b>11,846.3</b> | <b>0.4482</b>           | <b>1.0322</b>                     | <b>16,621.69</b>                     |
| 10       | 11,139.2        | 0.4363                  | 0.9707                            | 15,994.37                            |
| 11       | 10,420.1        | 0.4409                  | 0.9468                            | 15,663.98                            |
| 12       | 9,893.0         | 0.4395                  | 0.9391                            | 15,192.00                            |
| 13       | 9,198.5         | 0.4461                  | 0.9253                            | 15,228.63                            |
| 14       | 8,660.9         | 0.4195                  | 0.9354                            | 15,120.41                            |
| 15       | 8,127.6         | 0.4233                  | 0.9438                            | 15,148.68                            |
| 16       | 7,696.5         | 0.3758                  | 1.0510                            | 15,079.57                            |
| 17       | 7,088.8         | 0.3846                  | 1.0127                            | 15,562.75                            |
| 18       | 6,564.3         | 0.3884                  | 1.0145                            | 16,005.42                            |
| 19       | 6,356.8         | 0.3869                  | 0.9934                            | 15,681.57                            |



Figure 9: Training-pool cluster sizes for the selected  $K = 9$  solution.

When a query employee is asked, the cluster to which he belongs is detected and a cosine k-NN model is fitted on the employees in the training pool of that cluster only, and the top-K most similar employees are retrieved from the cluster. This is a hybrid a model-based unsupervised step builds up a candidate pool, and a memory-based neighbourhood step does the final ranking in this candidate pool.

## PART 3: EVALUATIONS

### 3.1 Overview

A strict offline protocol was developed to ensure the non-circular and fair comparison. The 50,000 employees were split 80/20 into a training pool of 40,000 and a held-out query set of 10,000. Both recommenders are installed on, and only use, the training pool, while the test employees are only used for issuing queries and scoring the returned ranking. PCA and KMeans are also fit on the training pool avoiding information leakage. Three common metrics were applied and then reused across both systems: Precision@K, MAP@K, and NDCG@K. All 10,000 query employees are taken out of the training set and the metrics are averaged over the test set.

### 3.2 Results

Table 3: Results of Both Models

| Model                           | Precision@K | MAP@K  | NDCG@K |
|---------------------------------|-------------|--------|--------|
| Content-Based Filtering (K = 5) | 0.4496      | 0.5161 | 0.5584 |
| Cluster-Based Hybrid (K = 13)   | 0.4494      | 0.5096 | 0.6042 |

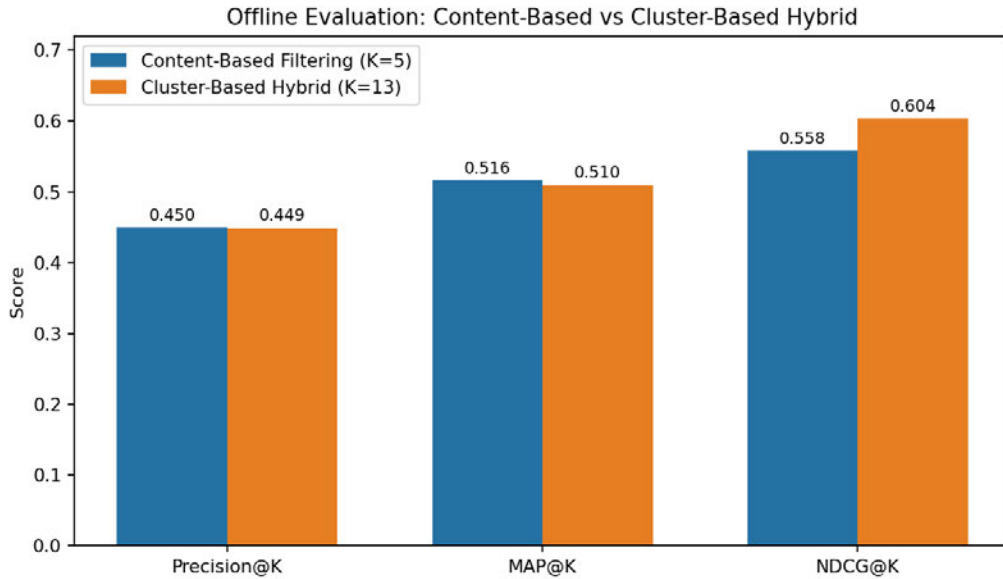


Figure 10: Metric comparison between the two recommenders.

Both systems markedly outperform the random-recommendation baseline (with about 45% of training employees being marked relevant; both have Precision around 0.45, and the story here is the ranking sensitive ones.) There is little difference between the systems with regard to Precision or MAP (0.4496 for the first and 0.4494 for the second; 0.5161 vs 0.5096 respectively), and both systems identify relevant employees with similar frequencies. The

Cluster Based Hybrid, however, has a significantly better NDCG of 0.6042 when compared with 0.5584 of the other algorithm, suggesting that within the wellness cluster, the similarity scores rank relevant employees higher than employees not relevant to that cluster in terms of ranking.

### **3.3 Cold Start Mitigations**

One of the main challenges in recommender systems is the Cold Start problem. This project addresses this challenge using a simple two-level approach. In the first case, when a new employee has no previous history but provides a wellness profile, the system compares that profile with existing employees using key features, such as sleep hours, weekly working hours, mental health, and job satisfaction. It then recommends employees with the most similar wellness profiles.

In the second case, when no information is available about a new employee, the system recommends employees with the highest overall wellness scores based on healthy sleep, good mental health, high job satisfaction, and balanced working hours. This provides a reasonable starting point until real employee data becomes available. The effectiveness of this approach was also tested using a synthetic employee profile generated by the VAE model. The system successfully returned relevant recommendations, showing that the proposed cold start strategy is practical and can support future employee onboarding.

## **PART 4: GENERATIVE AI ANALYSIS**

### **6.1 Role of GANs in Recommender Systems**

A Generative Adversarial Network is a training process that trains a Generator against a Discriminator, with the goal of creating profiles that are statistically indistinguishable from real employees when the Generator and Discriminator are in equilibrium. GANs have been applied in recommender systems (e.g., IRGAN, CFGAN) to supplement limited interaction data, create plausible what-if user profiles, and to generate hard adversarial negative samples to refine ranking models. The objective of this work is to use a GAN to create realistic records of employee wellness for expanding under-represented populations (e.g., severe-burnout profiles), stress test the recommenders, and create privacy-preserving synthetic populations for demos while not revealing any real employee data. The disadvantages are well understood: adversarial training is unstable on small tabular data sets, easily prone to mode collapse and has no explicit likelihood or structured latent space.

### **6.2 Role of VAEs in Wellness VAE**

Variational Autoencoders are trained to learn a latent representation, which is assumed to be drawn from a distribution, by encoding an input into the distribution and sampling from this distribution, then decoding back to the input, minimizing reconstruction loss and a KL-divergence regulariser. Mult-VAE and its successors are among the best collaborative-filtering models for recommendation research, and a VAE's smooth latent space also leads to controlled generation and interpolation between profiles.

This project doesn't discuss generative models on the abstract level, but rather brings a compact VAE into reality, specifically from PyTorch: WellnessVAE: a 6-16-3 encoder generating a 3d latent space, mirrored decoder with sigmoid outputs, trained over the MinMax scaled training pool, with average loss dropping from 0.6303 at epoch 20 to 0.5494 at epoch 60, Figure 12. The sampling from the latent prior and the inverse-scaling produced 5 synthetic employee profiles with realistic values, such as 43.5 weekly hours, 6.6 hours of sleep, mental-health around 90, satisfaction around 9.4. To close the loop between the generative analysis and the implemented system, one synthetic profile was fed into the cold-start recommender system as such as a genuinely new employee's profile. It pulled out five real training employees with similar sleep times (6.2-6.6 hours), workloads (42.7-44.1 hours), and mental-health values (89-91); this shows that the generated profiles can be directly used by the existing pipeline (no code changes needed), and that a VAE is a low friction method for stress testing or extending the cold-start pathway with more new, synthetic employees than the real data set contains.



Figure 11: WellnessVAE training loss (reconstruction + KL) over 60 epochs.

### 6.3 Generative versus Traditional Approaches for This System

Table 4 : Comparison of the implemented traditional systems with generative enhancements

| Aspect           | Traditional (CBF / Cluster-Hybrid)                                    | Generative (GAN / VAE)                                                                     |
|------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Data requirement | Works directly on observed profiles; no training beyond k-NN / KMeans | Needs sufficient data and tuning to learn the profile distribution                         |
| Cold start       | Handled by profile distance and population fallback                   | Can synthesise unlimited plausible new-employee profiles for testing and augmentation      |
| Explainability   | High: similar features are directly inspectable                       | Lower: latent dimensions are not directly interpretable                                    |
| Ranking quality  | Strong here (NDCG up to 0.60)                                         | Mult-VAE-style models could learn non-linear preference structure beyond cosine similarity |
| Risk             | Limited to linear/metric similarity                                   | Mode collapse (GAN), blurry averages (VAE), unrealistic feature combinations               |
| Compute          | Light; runs on CPU in seconds                                         | Heavier; benefits from GPU and larger data                                                 |

The limitations observed were brought up with candor, with just 60 epochs and a deliberately tiny network, the reconstruction is approximate, and before a production deployment would be used anything other than cold start stress-testing, more capacity, more data and validation that synthetic profiles don't encode unrealistic combinations of features would be needed. To directly compare the NDCG between three different recommenders: the cold-start heuristic, conditional VAE for generating an expected wellness profile, and Mult-VAE-style ranking head, future work is suggested.

## PART 5: CONCLUSION

Two different recommenders, Content Based Filtering and a KMeans Cluster Based Hybrid were designed on a common canonical feature pipeline and evaluated on a constant evaluation protocol in a non-circular setting on 10,000 questions held out. Both had good retrieval performance, Precision@K  $\sim 0.45$ , MAP  $\sim 0.51$ . Cluster-Based Hybrid outperformed on the ranking quality, NDCG 0.6042 vs 0.5584 and searched only a small fraction  $\sim 1/9$  of the number of candidates per query. The two-tiered strategy was implemented to combat the Cold-Start challenge and implemented end-to-end (EE) as a structural approach to counteract Scalability. Lastly, the Generative AI analysis didn't stop at the word; it actually introduced a PyTorch VAE model and generated synthetic profiles, which were directly input into the cold-start pathway to offer a real, critically assessed perspective on the potential improvements that generative models could bring to the existing cold-start systems developed here.

## Appendix A: Deliverables Mapping

Table 5: Traceability of the CA ONE assessment criteria to this report and the notebook.

| CA ONE Requirement                                                                                  | Where Addressed                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part 1: Two distinct recommender systems, rationale, pre-processing, feature engineering            | Part 2 in the PDF; Jupyter notebook cells: cleaning, EDA, ColumnTransformer pipeline, CBF, Cluster-Based Hybrid.                                                                                                                                                                                                                                                  |
| Part 2: Offline evaluation with NDCG and MAP; Cold-Start mitigation with demonstrated effectiveness | Part 3 in the PDF; Jupyter notebook cells: Precision@K / MAP@K / NDCG@K functions, head-to-head comparison, two-tier cold-start implementation                                                                                                                                                                                                                    |
| Part 3: Generative AI (GANs, VAEs) analysis and comparison to traditional methods                   | Section 4; notebook: WellnessVAE (PyTorch), synthetic profile generation, integration with cold-start                                                                                                                                                                                                                                                             |
| Code Repository / Python Notebook                                                                   | CA1_Human_Wellness_Recommender.ipynb<br>Git Hur Repository Link:<br><a href="https://github.com/taimoor0078/Employee-Wellness-Recommendation-System-Using-Content-Based-and-Cluster-Based-Hybrid-Recommendations.git">https://github.com/taimoor0078/Employee-Wellness-Recommendation-System-Using-Content-Based-and-Cluster-Based-Hybrid-Recommendations.git</a> |
| Group Presentation                                                                                  | Slides summarizing all parts                                                                                                                                                                                                                                                                                                                                      |